

EDMS 657 HW3

Alex Caramanico

2025-11-09

due 11/11

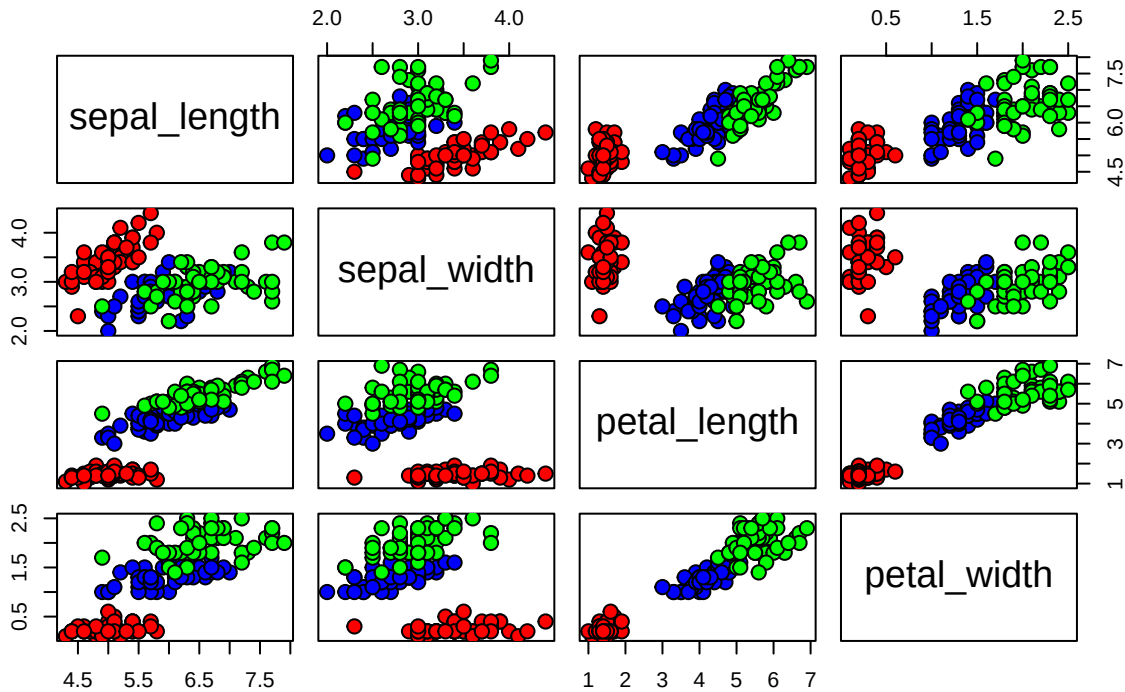
Cluster Analysis

1.

What is the natural modality of this dataset? In other words, does there appear to be more than one natural cluster in these data? Explain.

```
library(tidyverse)
library(ggplot2)
library(cowplot)
library(pvclust)
library(fpc)
library(psych)
library(factoextra)
iris <- read_table("C:/Users/jacma/OneDrive/school work/UMD/EDMS 657/Iris-Data.dat",
                  col_names = c("sepal_length", "sepal_width",
                                "petal_length", "petal_width",
                                "species"), skip=1)
iris$species <- as.factor(iris$species)
colors <- c("1" = "red", "2" = "blue", "3" = "green")
pairs(iris[, 1:4], pch = 21, cex = 1.5,
      bg = colors[iris$species], main = "Pairs Plot Colored by Species")
```

Pairs Plot Colored by Species



```
rm(colors)
```

```
iris %>%
  group_by(species) %>%
  summarize(mean_petal_l=mean(petal_length),
             mean_petal_w=mean(petal_width),
             mean_sepal_l=mean(sepal_length),
             mean_sepal_w=mean(sepal_width))
```

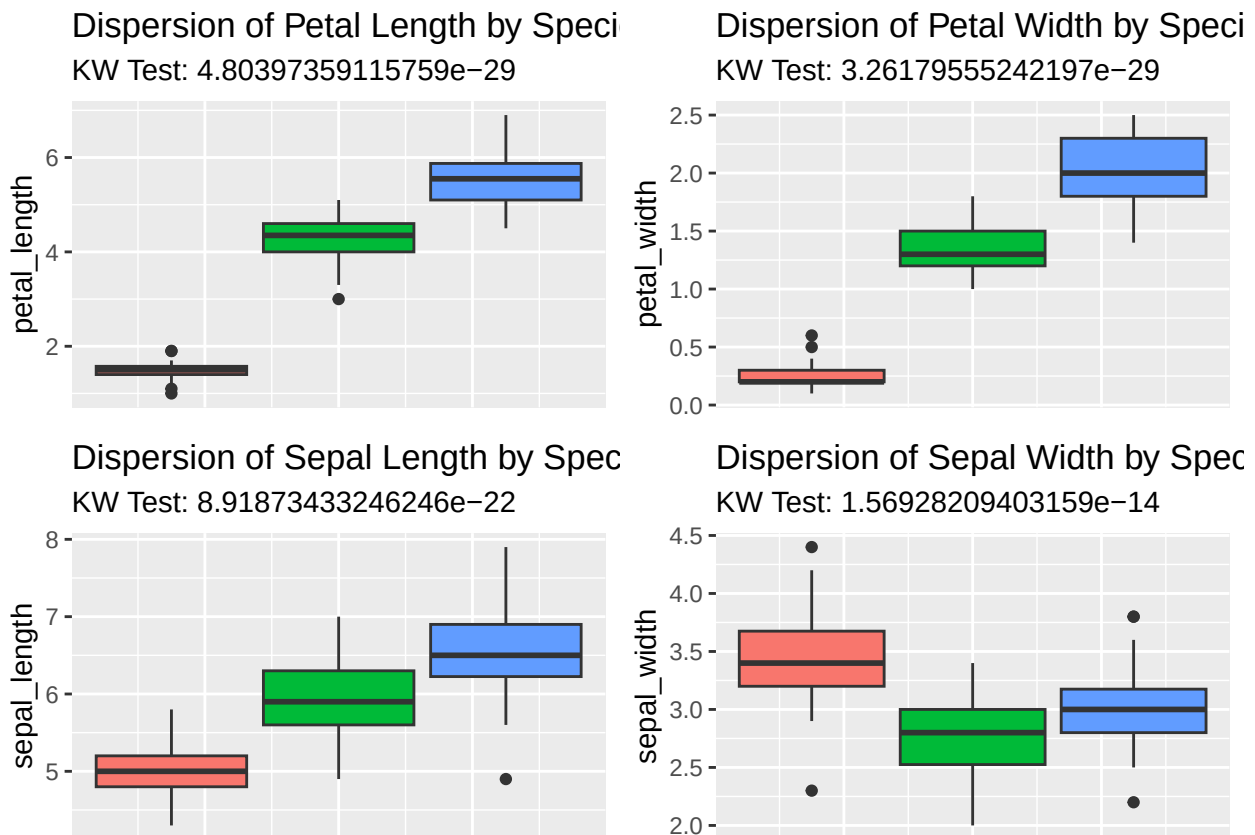
```
## # A tibble: 3 x 5
##   species mean_petal_l mean_petal_w mean_sepal_l mean_sepal_w
##   <fct>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 1          1.46        0.246        5.01        3.43
## 2 2          4.26        1.33         5.94        2.77
## 3 3          5.55        2.03         6.59        2.97
```

```
plot_grid(ggplot(iris) +
  geom_boxplot(mapping=aes(y=petal_length,group=species,fill=species)) +
  theme(axis.text.x = element_blank(),axis.ticks.x = element_blank(),
        legend.position = 'none') +
  labs(title="Dispersion of Petal Length by Species",
        subtitle=paste("KW Test:",kruskal.test(petal_length~species,data=iris)$p.value)),
  ggplot(iris) +
  geom_boxplot(mapping=aes(y=petal_width,group=species,fill=species)) +
  theme(axis.text.x = element_blank(),axis.ticks.x = element_blank(),
```

```

    legend.position = 'none') +
  labs(title="Dispersion of Petal Width by Species",
    subtitle=paste("KW Test:",kruskal.test(petal_width~species,data=iris)$p.value)),
  ggplot(iris) +
  geom_boxplot(mapping=aes(y=sepal_length,group=species,fill=species)) +
  theme(axis.text.x = element_blank(),axis.ticks.x = element_blank(),
    legend.position = 'none') +
  labs(title="Dispersion of Sepal Length by Species",
    subtitle=paste("KW Test:",kruskal.test(sepal_length~species,data=iris)$p.value)),
  ggplot(iris) +
  geom_boxplot(mapping=aes(y=sepal_width,group=species,fill=species)) +
  theme(axis.text.x = element_blank(),axis.ticks.x = element_blank(),
    legend.position = 'none') +
  labs(title="Dispersion of Sepal Width by Species",
    subtitle=paste("KW Test:",kruskal.test(sepal_width~species,data=iris)$p.value)),
  nrow=2)

```



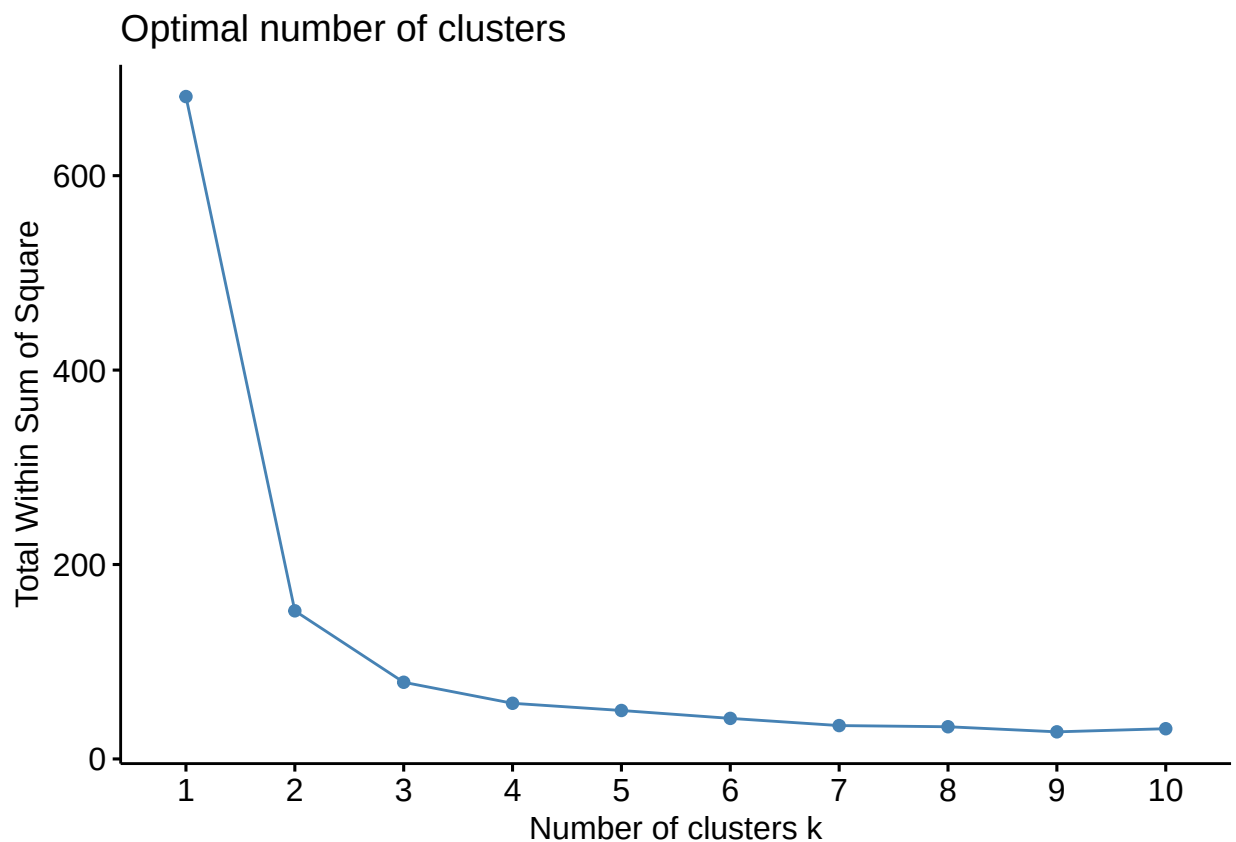
There are clearly at least two clusters in the data set, as the red species is quite far away from the blue and green species (in most of the pairs plots, best seen in those comparing petal measurements). Notably, this separation could be seen *without a color encoding* - there would be two separate grey-colored groups. Given the colorization by species, the difference between the blue group and the green group is less clear. One could envision a line or a border separating the two, but it may be difficult to do so without any overlap. See sepal_length vs sepal_width for example - the groups are fairly mixed together. The distinction between the groups is clearer in other plots, like petal_length vs sepal_length or petal_width. There are certainly at least two clusters in the data, and debatably three.

The summary table affirms these differences, where we can see large differences in the mean petal lengths and widths between species. Sepal length and width are more difficult to ascertain a difference, but the box plots show clear separations in the medians of each variable (although there is some overlap between groups for the sepal metrics). The Kruskal-Wallis tests (whose results were pasted in the subtitle of the respective plots, with the null hypothesis representing no difference between groups) show statistically significant differences in the medians between groups for *every* variable, which also suggests distinct clustering of data according to species. While this result does not tell us about the number of clusters, it tells us there are at least two of them.

2.

Using an appropriate clustering method, find a three-cluster solution for the iris data. State and justify the methods that you use. Report your findings.

```
# kmeans  
fviz_nbclust(iris[,1:4], kmeans, method = "wss") # 2 or 3 clusters
```



```
three_clust <- kmeans(iris[,1:4], centers = 3, nstart = 25)  
three_clust$size # original data was 50/50/50 - group 1 underrepresented, group 2 over
```

```
## [1] 38 62 50
```

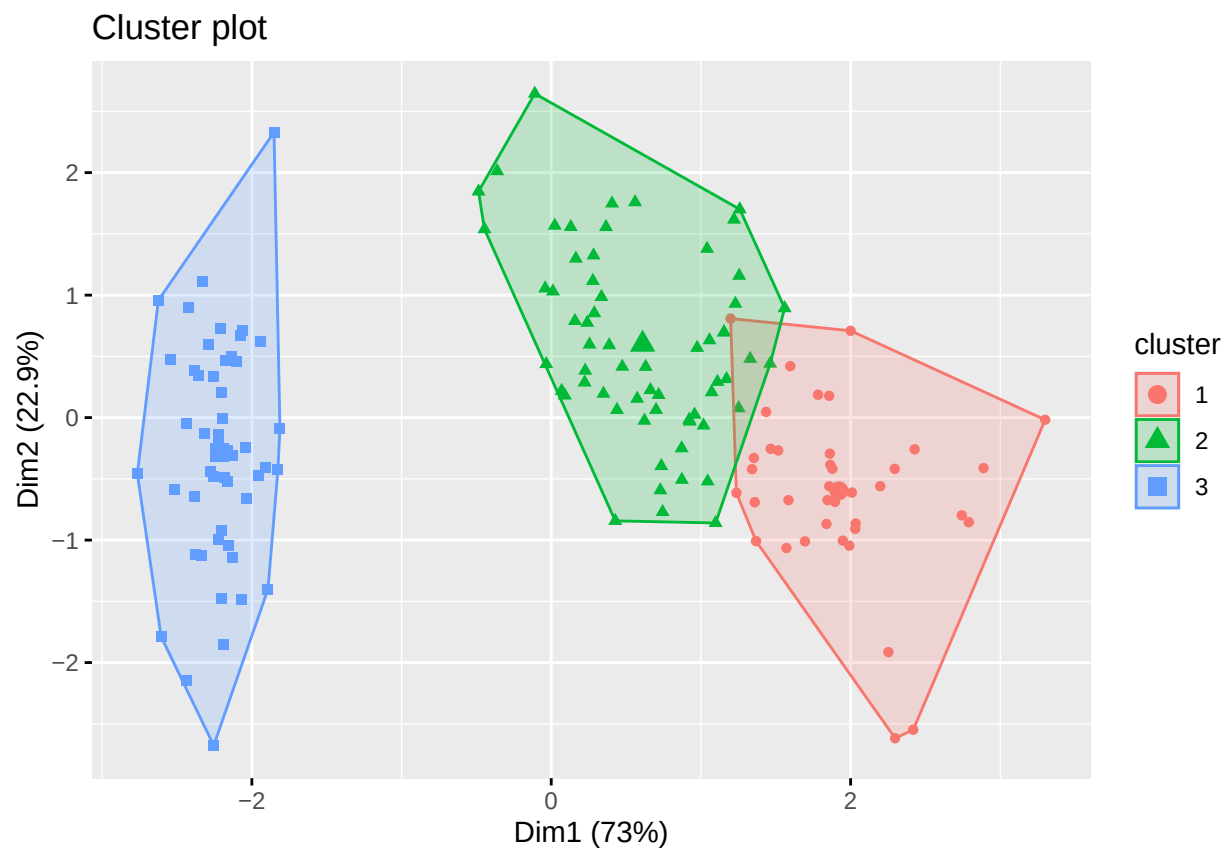
```
three_clust$centers # almost exactly the same as summary table
```

```
##      sepal_length sepal_width petal_length petal_width
## 1      6.850000     3.073684     5.742105     2.071053
## 2      5.901613     2.748387     4.393548     1.433871
## 3      5.006000     3.428000     1.462000     0.246000
```

```
three_clust$withinss # group 1 and 2 have larger errors
```

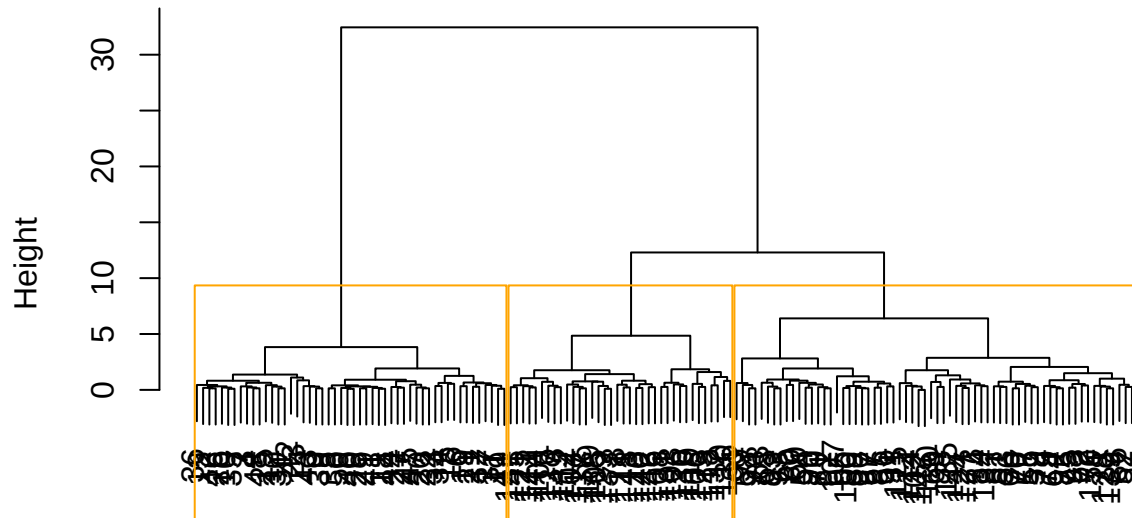
```
## [1] 23.87947 39.82097 15.15100
```

```
factoextra::fviz_cluster(three_clust, iris[,1:4], geom = "point")
```



```
# hierarchical
d <- dist(iris[,1:4], method = "euclidean")
fit <- hclust(d, method="ward.D2")
plot(fit) # display dendrogram
rect.hclust(fit, k=3, border="orange")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
rm(d)
```

Both k-means and hierarchical clustering methods were implemented above, but the former seems much more sensible for this data set. k-means is immediately motivated by the fact we have a solid idea on how many clusters there are in the data set, two or three. Hierarchical models do not require the analyst to specify the number of clusters, which could be useful for certain data sets. Another consideration is data set size, with k-means being preferred for large amounts of data due to the speed of its computations (the data set here is not very large, but storing a 150x150 distance matrix would not be preferred if its performance is not significantly better than the less-expensive k-means approach, see the following question). It is arguable whether the nesting of clusters is important, since the data is based on flower species. The analyst may be interested in the biological relationship between or “closeness” of the species, which can be implied by a hierarchical clustering result. All factors considered, hierarchical clustering is probably the most adequate due to the reasonable sample size, lack of a cluster specification requirement, and its interpretability in the problem context. On this last justification - we see from the dendrogram (when drawing rectangles that make three groups that two clusters are closer together than the other - they merge at a height around 12, whereas the last merge occurs all the way at a height over 30. This matches the intuition discussed in the first question, where the “blue” and “green” species were closer together in space than either of them were to the “red” species.

3.

How does your clustering solution correspond to the actual species identified in the data set? Use a classification table in your response.

```

# pretty sure the kmeans reversed the numbers for the groups, fix
for (i in 1:length(three_clust$cluster)){
  # did this in a for loop because my vector solution effectively removed a class
  if (three_clust$cluster[i]==1) three_clust$cluster[i] <- 3
  else if(three_clust$cluster[i]==3) three_clust$cluster[i] <- 1
}
all_results <- data.frame(kmeans=three_clust$cluster,
                          hier=cutree(fit, k=3),
                          actual=iris$species)
xtabs(~kmeans+actual,data=all_results)

```

```

##      actual
## kmeans  1  2  3
##      1 50  0  0
##      2  0 48 14
##      3  0  2 36

```

```

xtabs(~hier+actual,data=all_results)

```

```

##      actual
## hier  1  2  3
##      1 50  0  0
##      2  0 49 15
##      3  0  1 35

```

```

rm(all_results,fit,iris,three_clust,i)

```

For the plots in question 1, the “red” species was 1, “blue” was 2, “green” was 3. We see that for both methods, groups 2 and 3 were the hardest to distinguish, which matches the findings from the data exploration. The blue and green groups were much less distinguishable than the red group, so it is intuitive that the algorithms made the most errors between these species. Both algorithms had 16 total errors, but perfectly identified the “red” (1) species. Since the performances were extremely similar, decisions on which approach to use would depend on scalability of the data (much more data would favor k-means), and certainty on the number of species (more uncertainty, especially if the data was altered to include several others, would favor hierarchical because the researcher does not have to specify the number of clusters desired). Of course, there is always a balance between these two factors, so both must always be considered in practice

Alex Caramanico
EDMS 657 HW 3 - Second Half
11/10/2025

1

There is a known measurement equivalency between inches and centimeters. That is, $mean(inches) = (2.54)mean(cm)$. Refit the 2-class mixture of univariate normal distributions and implement this constraint into the model. Compare the fit of the model with the 2-class unconstrained univariate mixture using the Bayesian Information Criteria (or BIC—where lower BIC values indicate a relatively better fitting model).

Mplus code:

```
None
TITLE:Univariate Mixture Modeling (HW3-1)
DATA:FILE = "C:\Users\jacma\OneDrive\school work\UMD\EDMS
657\ME-Data-Mplus.dat";
VARIABLE:
    NAMES = ID sex age y;
    MISSING ARE ALL (-99);
    USEVARIABLES ARE y;
    CLASSES = c(2);
ANALYSIS:
    TYPE = MIXTURE; !accelerated EM is default
    STARTS = 50 10;
    STSCALE = 5; !1 to 9, perturbation "size" to start at different places
    STITERATIONS = 50; ITERATIONS = 1000;
    SDITERATIONS = 250; MITERATIONS = 500;
    MCONVERGENCE = 1E-5;

MODEL:
%overall%
y*; [y*];
%c#1% !class 1 is inches
y*5;[y*18](mu1); !name this as mean 1
%c#2%
y*15; [y*45](mu2); !name this as mean 2
MODEL CONSTRAINT: mu2 = 2.54*mu1;

OUTPUT: TECH1 TECH4; !TECH1: parameters being estimated, TECH8: iter history

SAVEDATA:
    FILE IS ME-Data-Mplus-2.dat;
    SAVE = CPROB; !save posterior probabilities, most likely class membership
```


2.

There is a known measurement equivalency between inches and centimeters. That is, $var(inches) = (2.54)^2 var(cm)$. Refit the 2-class mixture of univariate normal distributions and implement this constraint into the model. Compare the fit of the model with the 2-class unconstrained univariate mixture using the Bayesian Information Criteria (or BIC—where lower BIC values indicate a relatively better fitting model).

Mplus code:

```
None
TITLE:Univariate Mixture Modeling (HW3-2)

DATA:FILE = "C:\Users\jacma\OneDrive\school work\UMD\EDMS
657\ME-Data-Mplus.dat";

VARIABLE:
    NAMES = ID sex age y;
    MISSING ARE ALL (-99);
    USEVARIABLES ARE y;
    CLASSES = c(2);

ANALYSIS:
    TYPE = MIXTURE;
    STARTS = 50 10;
    STSCALE = 5;
    STITERATIONS = 50; ITERATIONS = 1000;
    SDITERATIONS = 250; MITERATIONS = 500;
    MCONVERGENCE = 1E-5;

MODEL:
%overall%
y*; [y*];
%c#1% !class 1 is inches!
y*5 (var1) ;[y*18](mu1); !name variance 1
%c#2%
y*15 (var2); [y*45](mu2); ! name variance 2
MODEL CONSTRAINT: var2 = 2.54^2*var1;

OUTPUT: TECH1 TECH4; !TECH1: parameters being estimated, TECH8: iter history

SAVEDATA:
    FILE IS ME-Data-Mplus-2.dat;
    SAVE = CPROB; !save posterior probabilities, most likely class membership
```

3.

Create a table of your results with rows as the 3 models being fit and columns representing parameters estimates and BIC value.

Model	μ_{in}	μ_{cm}	σ_{in}^2	σ_{cm}^2	ϕ_{cm}	BIC	BIC (n-adjusted)
2-class Unconstrained	18.25	45.66	5.66	14.92	.757	13041.47	13025.58
Mean constrained	18.01	44.74 (= $2.54\mu_{in}$)	5.63	15.01	.757	13039.55	13026.84
Variance Constrained	18.16	45.59	3.05	19.66 (= $(2.54)^2\sigma_{in}^2$)	.759	13152.29	13139.5

Note that the values for μ_{cm} and σ_{cm}^2 for the mean-constrained and variance-constrained models respectively were not explicitly estimated by the model, and thus are not “parameters”. Their values were computed for the table based on the conversion used in the actual constraining of the respective models, as shown by the expressions in the parentheses.

Based on the results, the researcher would probably prefer the mean-constrained or unconstrained model because they have the lowest BIC and adjusted BIC, respectively. Including constraints on the parameters makes intuitive sense, since the relationship/conversion between centimeters and inches is known. If it was suspected that the data contained two different measurements, then the constraints are a natural inclusion for the models. It may be even better to test a model that constrains both the mean *and* the variance parameters for centimeters. Overall, the parameter estimates are extremely similar across each model (notably the means and the mixture proportions). However, the variances in the constrained models are very different compared to the other models, with the variance in inches being much lower and the variance in centimeters being much higher than the others in their respective columns.